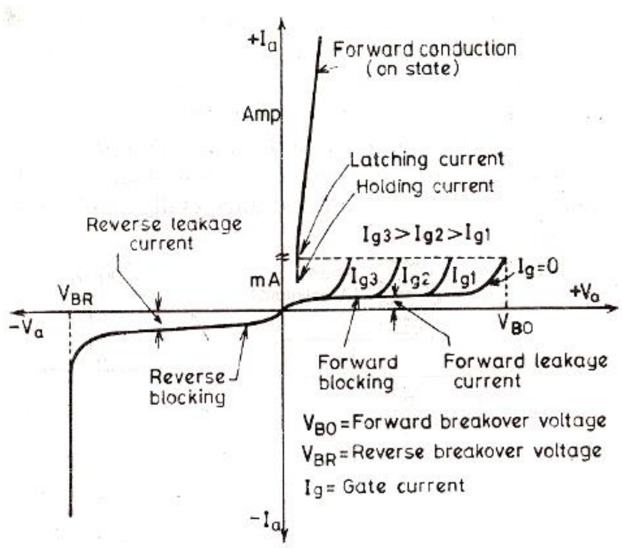
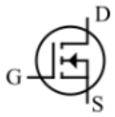
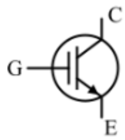
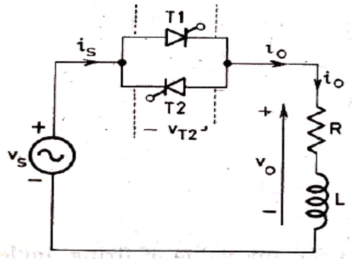
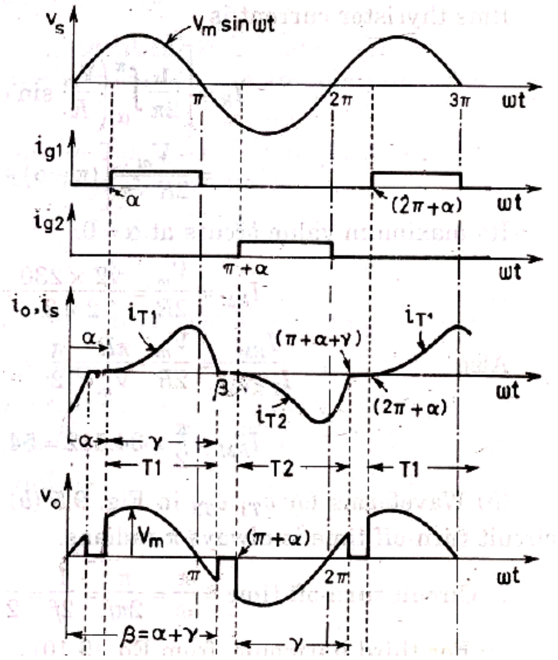
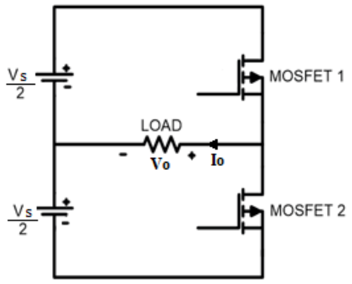
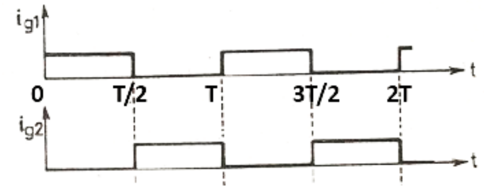
Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY B.TECH DEGREE S6 (R,S) EXAMINATION JUNE 2023 (2019 SCHEME)		
Course Code: EET306		
Course Name: POWER ELECTRONICS		
PART A		
		Answer all questions. Each question carries 3 marks
	Q1	Sketch the VI characteristics of SCR and explain the importance of latching current in turn on process of SCR.
	Ans	 **Latching Current** As the thyristor moves from forward-blocking to forward conduction, the external circuit must allow to flow a sufficient anode current to keep the device in ON state. **Minimum value of anode current to maintain the SCR in ON state even though the gate signal is removed.**
	Q2	Compare the characteristic features of MOSFET and IGBT.

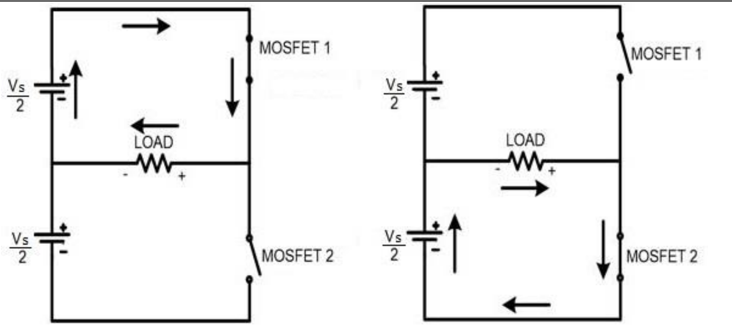
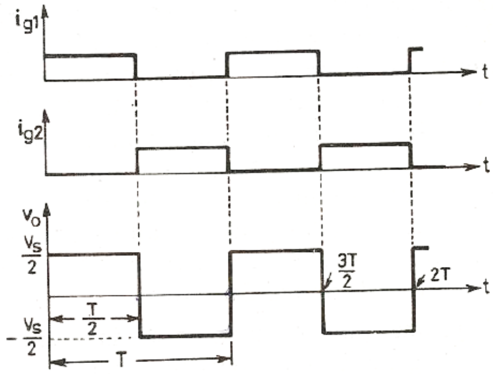
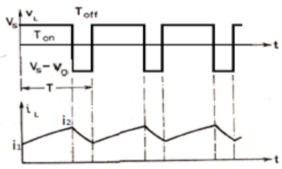
Ans

Power MOSFET	**IGBT**
Metal Oxide Semiconductor Field Effect Transistor	Insulated Gate Bipolar Transistor
Three terminal semiconductor device. Three terminals are Gate, Drain and Source	Three terminal semiconductor device. Three terminals are Gate, Emitter and Collector

MOSFET is capable of handling only low to medium voltage and power.	IGBT has ability to handle very high voltage and high power.
MOSFET can be used for very high frequency (of the order of MHz) applications.	IGBT can only be used for relatively low frequencies, up to a few kHz.
MOSFET produces higher forward voltage drop than IGBT.	When IGBT is conducting current, it produces comparatively low forward voltage drop.
	

Q3	Explain the advantage of freewheeling diode in halfwave converter, when the load is inductive in nature.
Ans	• Current Path: The freewheeling diode, also known as the commutation diode or freewheeling rectifier, offers a low-resistance path for the inductive load's current when the SCR is turned off. This allows the inductive load to continue to circulate current in a controlled manner, preventing abrupt voltage spikes and ensuring a smooth transition. • Reduces Stress on SCR: By providing a path for the inductive load current, the freewheeling diode helps protect the SCR from voltage spikes and excessive reverse voltage. This extends the lifespan of the SCR and improves the overall reliability of the converter. • Energy Recovery: The freewheeling diode also allows energy to be recovered from the inductive load during the off-period of the SCR. This energy can be used to power other parts of the circuit or to reduce power losses, making the converter more efficient.
Q4	A single phase full bridge converter, connected to 230 V, 50Hz source is feeding a load having resistance R=10Ω with a large inductance that makes the load current ripple free and continuous. Calculate the average value of load current for a firing angle of 60˚.
Ans	Large value of inductor $\Rightarrow$ continuous conduction. $V_0 = \frac{2V_m}{\pi}\cos\alpha$. $= \frac{2\times 230\times\sqrt{2}}{\pi}\times\cos(60) = 103.536\text{ V}$ $\therefore$ Average load current $= \frac{V_0}{R} = \frac{103.536}{10} = \underline{\underline{10.35\text{ A}}}$
Q5	Explain the working of single phase full wave AC voltage controller with RL load with

		neat circuit diagram, output voltage and current waveform.
	Ans	 
	Q6	Sketch the circuit diagram and the output voltage waveform of a single phase half-bridge Voltage Source Inverters with R load and explain its working.
	Ans	  During 0 – T/2 During T/2 – T

		 
	Q7	Derive the expression for output voltage of step up chopper in terms of input voltage and duty cycle.
	Ans	**Average Output voltage** Energy stored = $V_s \times \frac{(i_1+i_2)}{2} \times T_{on}$  Energy released = $(V_s - V_o) \times \frac{(i_1+i_2)}{2} \times T_{off}$ $V_s \times \frac{(i_1+i_2)}{2} \times T_{on} + (V_s - V_o) \times \frac{(i_1+i_2)}{2} \times T_{off} = 0$

<table>
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$$V_s \times \frac{(i_1+i_2)}{2} \times T_{on} + (V_s - V_o) \times \frac{(i_1+i_2)}{2} \times T_{off} = 0$$

$$V_s \times T_{on} + (V_s - V_o) \times T_{off} = 0$$

$$V_sT_{on} + V_sT_{off} - V_oT_{off} = 0$$

$$V_s(T_{on} + T_{off}) - V_oT_{off} = 0$$

$$V_o = V_s\frac{(T_{on} + T_{off})}{T_{off}} = V_s\left(\frac{T_{on}}{T_{off}} + 1\right)$$

$$V_o = V_s\left(\frac{T_{on}}{T - T_{on}} + 1\right)$$

$$V_o = V_s\left(\frac{1}{\frac{T - T_{on}}{T_{on}}} + 1\right) = V_s\left(\frac{1}{\frac{T}{T_{on}} - 1} + 1\right)$$

$$V_o = V_s\left(\frac{1}{\frac{1}{D} - 1} + 1\right) = V_s\left(\frac{D}{1 - D} + 1\right)$$

$$\boxed{V_o = V_s\left(\frac{D + 1 - D}{1 - D}\right) = \frac{V_s}{1 - D}}$$

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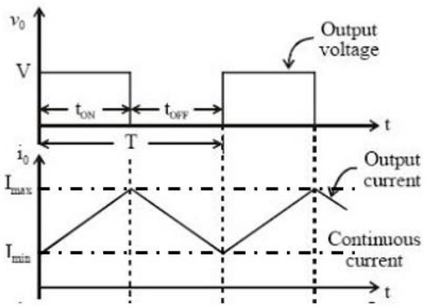
<tr><td></td><td>Q8</td><td>Explain the current limit control in DC-DC converter circuits.</td></tr>
<tr><td></td><td>Ans</td><td>

<u>Current Limit Control</u>

- In this control strategy, a specific limit is applied on the current variation
- Current is allowed to fluctuate or change only between 2 values i.e. maximum current (I_{max}) and minimum current (I_{min}).
- When the current is at minimum value, the chopper is switched ON.
- After this instance, the current starts increasing, and when it reaches up to maximum value, the chopper is switched off allowing the current to fall back to minimum value.



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<tr><td></td><td>Q9</td><td>List the advantages of Electric Drives.</td></tr>
<tr><td></td><td>Ans</td><td>

- Energy Efficiency: Electric drives are highly efficient, with the ability to convert electrical energy into mechanical work with minimal losses. This efficiency helps reduce energy consumption and operating costs.
- Precise Control: Electric drives provide precise control over speed, torque, and position, making them suitable for applications that require accuracy and consistency, such as robotics and CNC machines.
- Variable Speed Operation: Electric drives can easily operate at variable speeds, allowing for better control of machinery and equipment. This capability is crucial in applications like pumps, fans, and conveyors.
- High Starting Torque: Electric drives can provide high starting torque, which is

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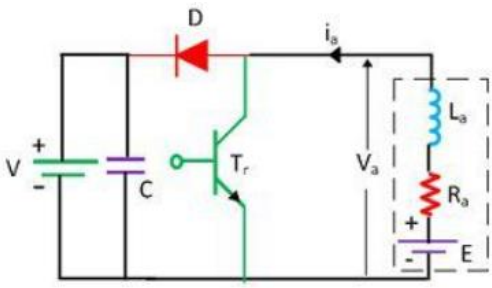
- useful in applications where equipment needs to overcome initial inertia or resistance, such as elevators and cranes.
- Reversible Operation: Electric drives can reverse the direction of rotation simply by changing the direction of current flow, making them versatile for bidirectional applications.
- Low Maintenance: Electric drives typically have fewer moving parts compared to mechanical drives, resulting in lower maintenance requirements and longer service life.
- Quiet Operation: Electric drives operate quietly, which is beneficial in environments where noise pollution is a concern, such as residential areas or hospitals.
- Reduced Emissions: Electric drives produce no direct emissions at the point of use, making them environmentally friendly and suitable for reducing greenhouse gas emissions when powered by renewable energy sources.
- Regenerative Braking: Some electric drives can recover energy during braking and feed it back into the electrical grid or use it to power other parts of the system, improving overall energy efficiency.
- Compact and Space-Efficient: Electric drives are often compact and can be easily integrated into existing systems, saving space in industrial settings.

Q10 Chopper for regenerative braking operation is shown in the figure below. The transistor Tr is operated periodically with a period T and on-period of ton. The waveform of motor terminal voltage va and armature current ia for continuous conduction is shown in the figure below. The external inductance is added to increase the value of La. When the transistor is on, ia increased from ia1 to ia2.



The mechanical energy is converted into electrical energy by the motor, now working as a generator, partly increased the stored magnetic energy in the armature circuit inductance and the remainder is dissipated in armature and transistors.

When the transistor is turned off, the armature current flows through diode D and the source V and reduces from ia2 to ia1. The stored electromagnetic energy and the energy supplied by the machine are fed to the source. The interval $0 \leq t \leq$ ton is called energy storage interval and the interval ton $\leq t \leq T$ called the duty interval.

PART B

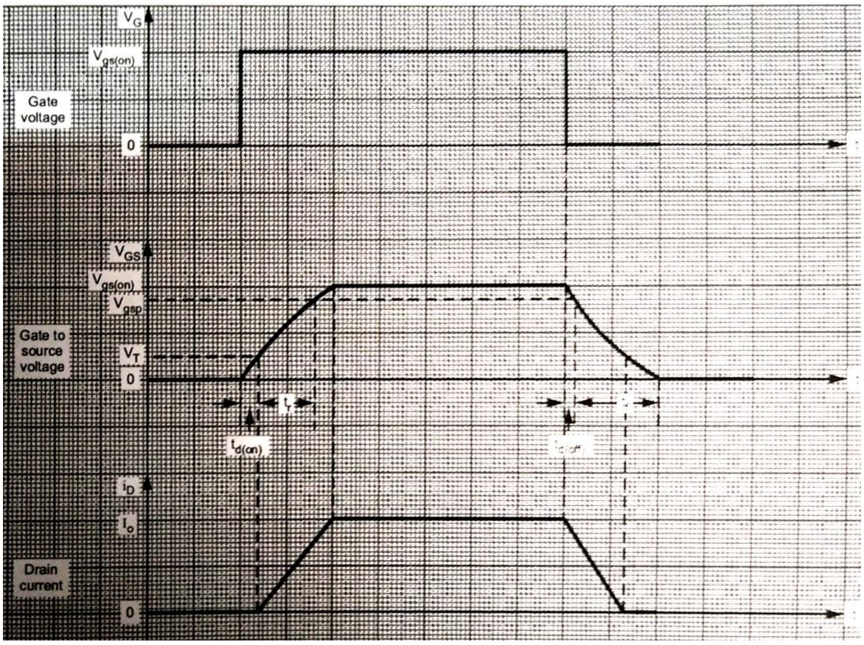
Answer any one full question from each module. Each question carries 14 marks

Module 1

Q11	a) Explain the switching characteristics of Power MOSFET with neat diagram (5 Mark) b) Illustrate the different turn on methods of SCR. (9 Marks)

Ans

a)



Turn-On Time: This is the time it takes for the MOSFET to transition from the OFF-state to the ON-state. It includes the delay time and the rise time.

- **Delay Time:** This is the time it takes from the application of the gate voltage until the MOSFET starts conducting. It is primarily determined by the gate capacitance and the gate resistor (if used). A smaller gate resistor reduces t_d but may increase gate current.
- **Rise Time:** This is the time taken for the MOSFET to fully turn ON once it starts conducting. It depends on the internal capacitances of the MOSFET and the gate voltage.

Turn-Off Time: This is the time it takes for the MOSFET to transition from the ON-state to the OFF-state. It includes the delay time and the fall time.

- **Delay Time:** This is the time it takes from the removal of the gate voltage until the MOSFET starts turning OFF. It is primarily determined by the gate resistor and the gate-source capacitance. A smaller gate resistor can reduce.
- **Fall Time:** This is the time taken for the MOSFET to fully turn OFF once it starts turning OFF. It depends on the internal capacitances and the gate voltage.

b)

- **Forward Voltage Triggering:** A forward voltage is applied across anode and cathode terminals with gate terminal open. As forward voltage is increased above forward break over voltage, avalanche breakdown will occur at junction J2. As other junctions J1 and J3 are already forward biased, breakdown of J2 allows free movement of charges across the junctions and as a result, large forward anode current flows.

- **Gate Triggering:** Turning on of SCR by gate triggering is simple, reliable and efficient, therefore it is the most usual method of firing the SCR. Here, for turning on the SCR a positive voltage is applied across gate and cathode terminals. Thus SCR can be turned on at a forward voltage less than the forward break over voltage.
- **dv/dt Triggering:** dv/dt Triggering is the technique in which SCR is turned ON by changing the forward bias voltage with respect to time. dv/dt itself means rate of change of voltage w.r.t time. Junction J2 is reversed biased in a forward blocking mode of SCR. A reversed biased junction may be treated as a capacitor due to presence of space charges in the vicinity of reversed biased junction. Thus the current through the reversed biased junction J2 is directly proportional to (dv/dt). Therefore if the rate of rise of forward voltage i.e. (dv/dt) is high, the charging current I will also be high. This charging current acts like gate current and turns ON the SCR even though the gate current is zero.
- **Temperature Triggering:** During forward blocking, most of the applied voltage appears across the reverse biased junction J2. This voltage across the junction associated with leakage current may raise the temperature of this junction. With increase in temperature, leakage current further increases. This cumulative process may turn on the SCR at some high temperature.
- **Light Triggering:** This triggering is only applicable for light activated SCRs (LASCR). Here, a pulse of light of appropriate wavelength is used for turning on SCR. When the intensity of this light exceeds a certain value, SCR will be turned on.

Q12

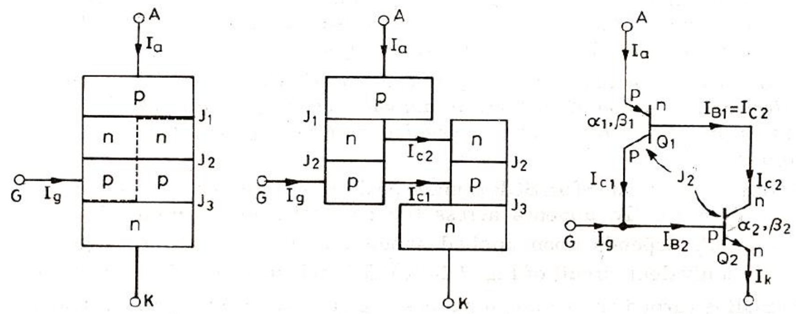
a) Explain the Two transistor analogy of SCR with significant equation. List the importance of current gain factor in turn on process. (7 Marks)

b) Illustrate di/dt protection and dv/dt protection of SCR and explain its significance (7 Marks)

Ans

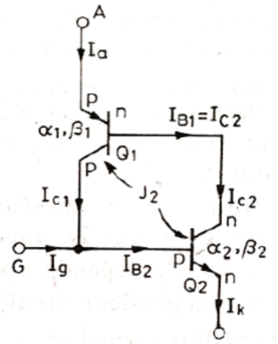
a)

Basic operating principle of SCR, can be easily understood by the two transistor model (or analogy) of SCR. If we bisect it through the dotted line, we will get two transistors.



- I_C is collector current, I_E is emitter current, I_{CBO} is forward leakage current, α is common base forward current gain and relationship between I_C and I_E is

$$I_C = \alpha I_E + I_{CBO}$$



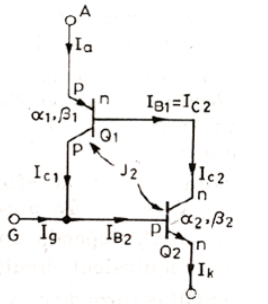
- For transistor Q1,

$$I_{C1} = \alpha_1 I_a + I_{CBO1}$$

- For transistor Q2,

$$I_{C2} = \alpha_2 I_k + I_{CBO2}$$

By applying KCL



$$I_a = I_{C1} + I_{C2}$$

$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 I_k + I_{CBO2}$$

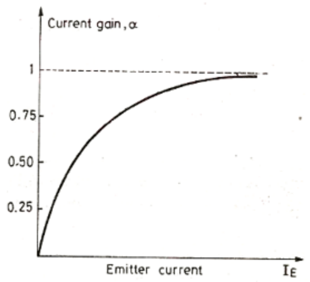
$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 (I_{B2} + I_{C2}) + I_{CBO2}$$

$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 (I_g + I_{C1} + I_{C2}) + I_{CBO2}$$

$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 (I_g + I_a) + I_{CBO2}$$

$$I_a = \frac{\alpha_2 I_g + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

$$I_a = \frac{\alpha_2 I_g + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$



- During forward blocking mode, only leakage current will flow through SCR and $I_g = 0$.
- Since anode current is small, α1 and α2 will be small
- Thus SCR remains in OFF state.
- When I_g is applied, I_a increases and thus α1 and α2 will increase.
- Again Ia increases.
- This cumulative process will turn on the SCR.

b) di/dt Protection

- When a thyristor is forward biased and is turned on by a gate pulse, conduction of anode current begins in the immediate neighbourhood of the gate-cathode junction.
- Thereafter, the current spreads across the whole area.
- If the rate of rise of anode current is higher than the spread velocity, local hot spots will be formed.
- This localised heating may destroy the thyristor.

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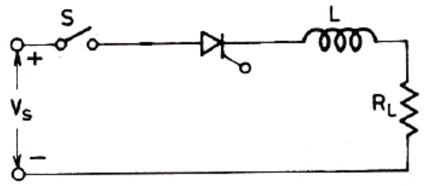
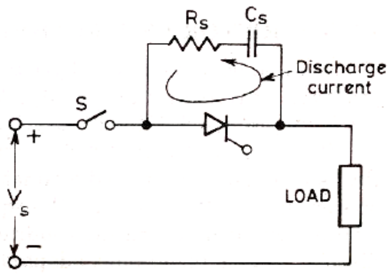
- Therefore, the rate of rise of anode current at the time of turn-ON must be kept below the specified limit.
- This can be achieved by connecting a small inductor in series with the thyristor.



dv/dt Protection

- With forward voltage across the anode and cathode of a thyristor, the two outer junctions are forward biased but the inner junction is reverse biased.
- This reverse biased junction act as a capacitor, with the entire anode to cathode voltage across it.
- Sudden rate of change of voltage, will lead to large charging current.
- This may damage the thyristor.
- Also this current plays the role of gate current and turn ON SCR, even though there is no gate signal (false triggering).
- Thus, rate of rise of forward voltage should be kept in specified limit.
- This can be achieved by connecting a capacitor in parallel with the thyristor.
- In a capacitor, voltage cannot be changed instantaneously.
- To limit the current during the discharging of the capacitor, a resistor is also connected in series.



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Module 2

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<tr><td></td><td>Q13</td><td>

a) Draw the circuit diagram of three phase fully controlled converter with RLE load. Derive the output equation assuming continuous conduction, ripple free operation (6 Marks)

b) Sketch and compare the output voltage waveforms of three phase fully controlled converter with RLE load for firing angles i) $\alpha = 0^\circ$ and ii) $\alpha = 60^\circ$. Show the conducting devices in each firing angle. (8 Marks)

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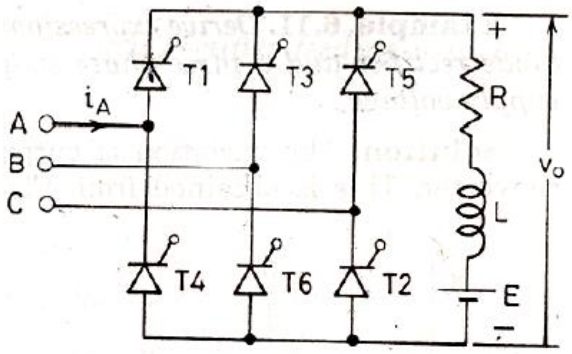
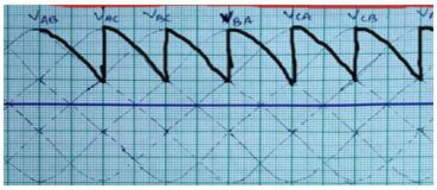


Average Output Voltage



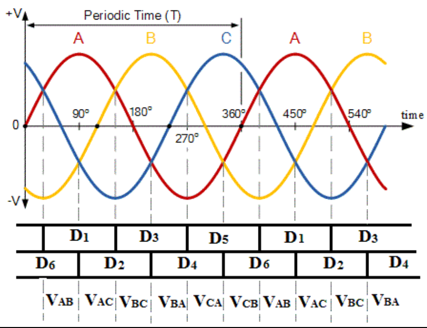
$$V_o = \frac{\int_{\alpha+30}^{\alpha+90} \sqrt{3}V_m \sin(\omega t + 30) d\omega t}{\pi/3}$$

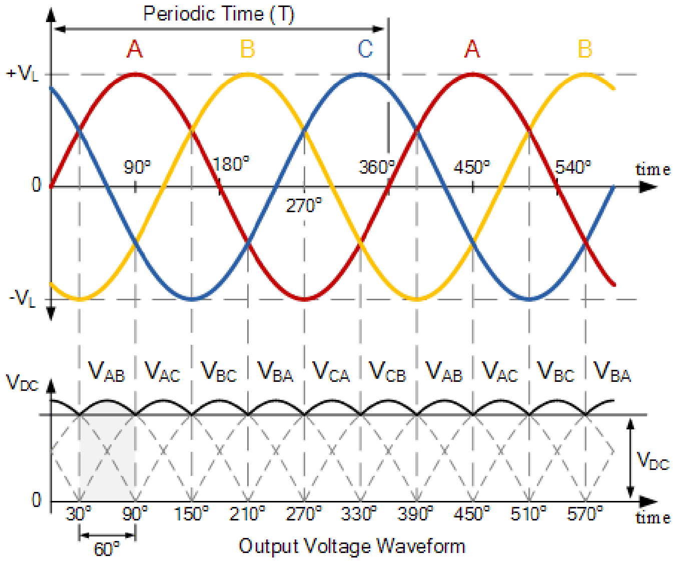
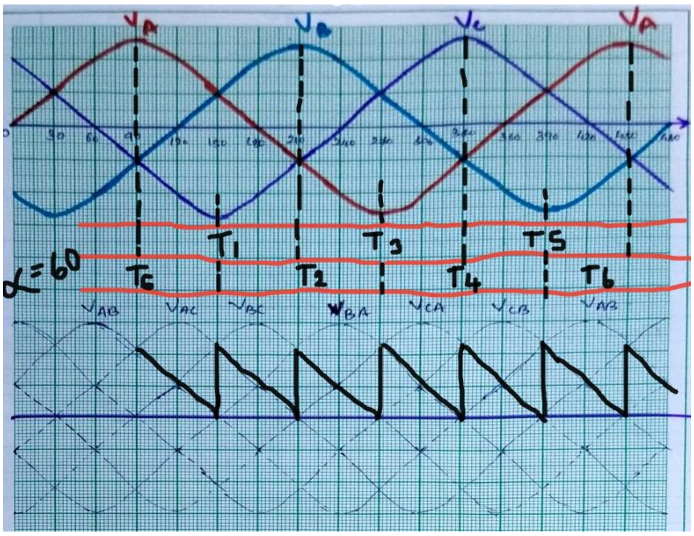
$$= \frac{3\sqrt{3}V_m}{\pi} \int_{\alpha+30}^{\alpha+90} \sin(\omega t + 30) d\omega t = \frac{3\sqrt{3}V_m}{\pi} \left[-\cos(\omega t + 30)\right]_{\alpha+30}^{\alpha+90}$$

$$\boxed{V_o = \frac{3\sqrt{3}V_m}{\pi} \cos\alpha}$$

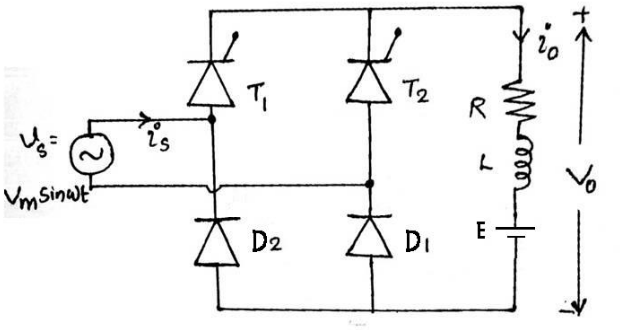
b)

Firing angle = 0 degree

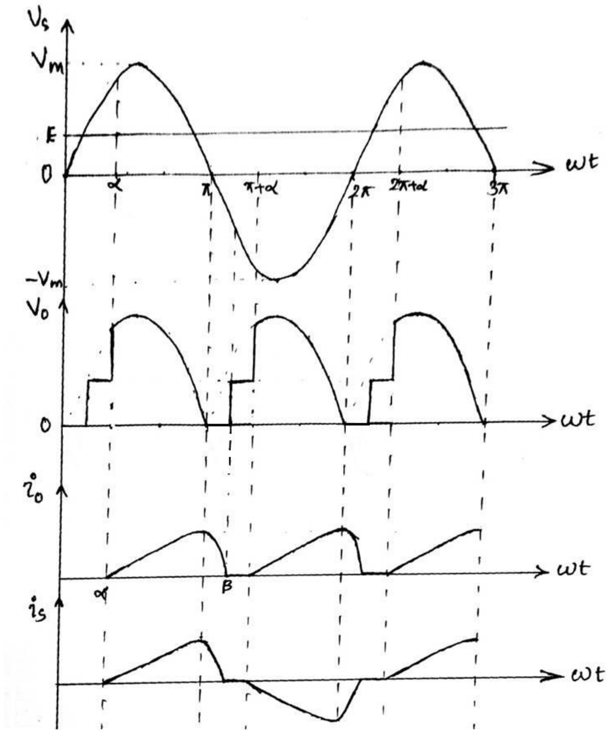


		 Firing angle = 60 degree 
	Q14	a) Illustrate the working of single phase half controlled bridge rectifier with RLE load and derive the output voltage equation in discontinuous conduction mode. Sketch the output voltage and output current waveform. (10 Marks) b) Compare the working of fully controlled converter and semi converter with voltage - firing angle plots. (4 Marks)

	Ans	a)



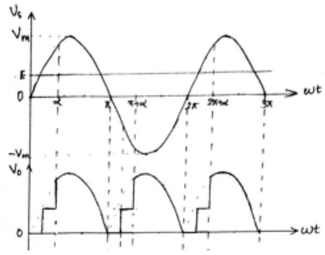
T1
T2
D2
D1
R
L
E
i_o
V_0
V_s = $V_m \sin\omega t$
i_s



V_s
V_m
E
0
α
π
$\pi+\alpha$
2π
$2\pi+\alpha$
3π
$-V_m$
V_0
i_0
B
i_s
ωt

Average of output voltage (discontinuous conduction)

$$V_o = \frac{\int_{\alpha}^{\pi} (V_m \sin \omega\, t) d\omega t + \int_{\beta}^{\pi+\alpha} E d\omega t}{\pi}$$

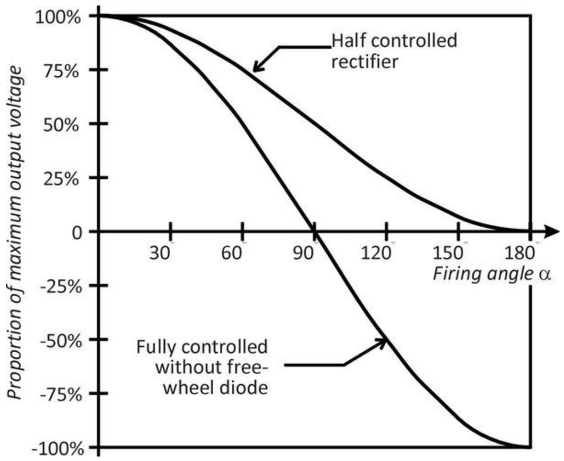


$$= \frac{V_m}{\pi} [-\cos \omega\, t]_{\alpha}^{\pi} + E[\omega t]_{\beta}^{\pi+\alpha}$$

$$= \frac{V_m}{\pi} [(-\cos \pi) - (-\cos \alpha)] + E[\pi + \alpha - \beta]$$

$$\boxed{V_o = \frac{V_m}{\pi} [\cos \alpha + 1] + E[\pi + \alpha - \beta]}$$

b)



Full Converter	Semi converter
Two quadrant converter	Single quadrant converter
Bidirectional power flow	Unidirectional power flow
Four SCRs	2 SCRs and 2 Diodes

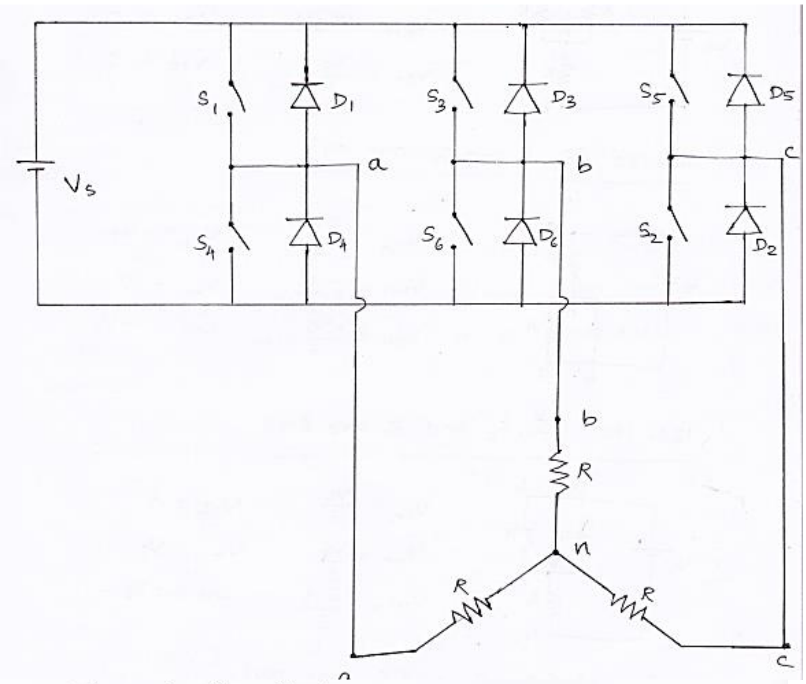
Module 3

Q15 a) Illustrate the working of three phase bridge inverter with R load with 180° conduction mode. Sketch the corresponding phase voltage waveforms and line voltage waveforms (10 Marks)

b) List out the merits and demerits of 120° conduction mode over 180° mode inverter. (4 Marks)

Ans

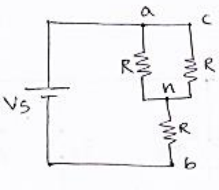
a)



* 180° Conduction Mode

In this mode, each device conducts for a duration of 180°. Switches are gated in a regular intervals of 60°, in a particular sequence.

0°–60° (S_5, S_6 and S_1 are ON)

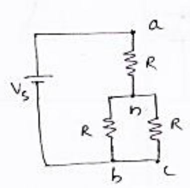


$V_{an} = \frac{V_s}{3}$ $\quad$ $V_{ab} = V_{an} - V_{bn} = V_s$

$V_{bn} = \frac{-2V_s}{3}$ $\quad$ $V_{bc} = -V_s$

$V_{cn} = \frac{V_s}{3}$ $\quad$ $V_{ca} = 0$

60°–120° (S_6, S_1 and S_2 are ON)

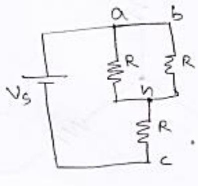


$V_{an} = \frac{2V_s}{3}$ $\quad$ $V_{ab} = V_s$

$V_{bn} = -\frac{V_s}{3}$ $\quad$ $V_{bc} = 0$

$V_{cn} = -\frac{V_s}{3}$ $\quad$ $V_{ca} = -V_s$

120°-180° (S_1, S_2 and S_3 are ON)

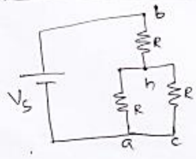


$V_{an} = \frac{V_s}{3}$ $\quad$ $V_{ab} = 0$

$V_{bn} = \frac{V_s}{3}$ $\quad$ $V_{bc} = V_s$

$V_{cn} = \frac{-2V_s}{3}$ $\quad$ $V_{ca} = -V_s$

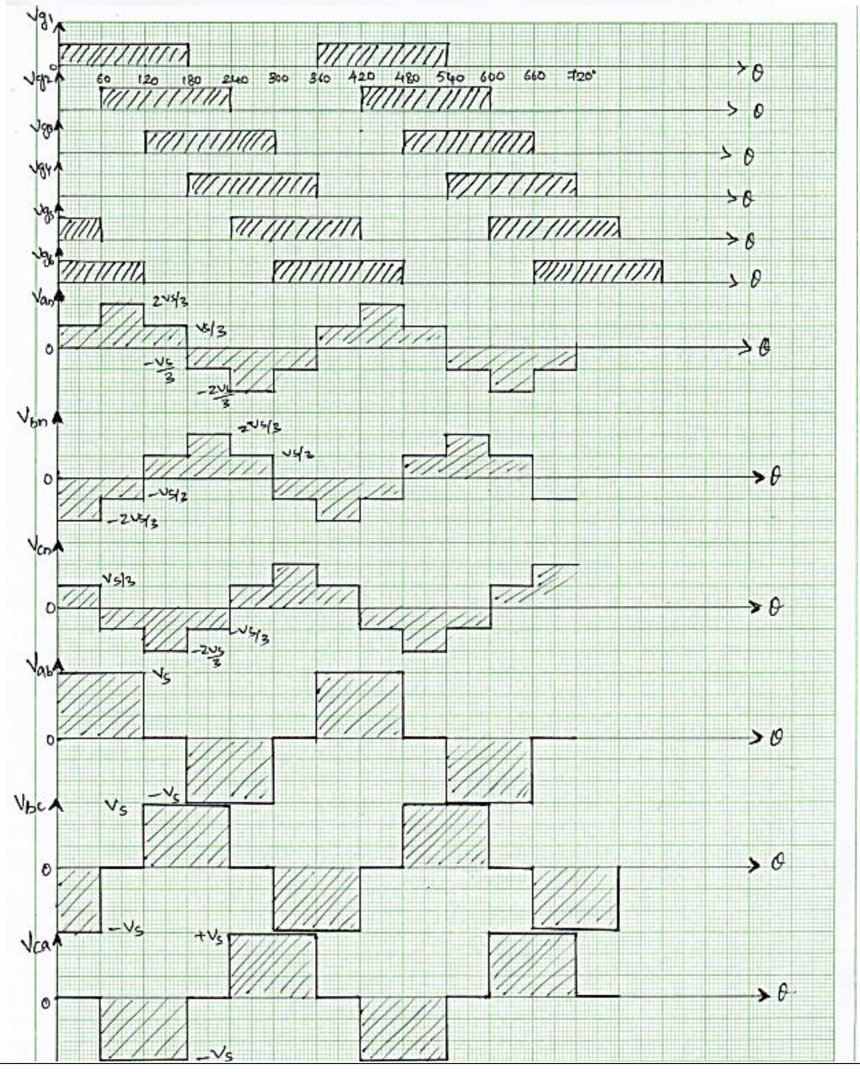
180°-240° (S_2, S_3 and S_4 are ON)



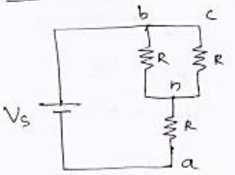
$V_{an} = -\frac{V_s}{3}$ $\quad$ $V_{ab} = -V_s$

$V_{bn} = \frac{2V_s}{3}$ $\quad$ $V_{bc} = V_s$

$V_{cn} = -\frac{V_s}{3}$ $\quad$ $V_{ca} = 0$

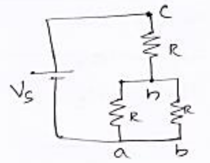


240° – 300° (S_3, S_4 and S_5 are ON)



$$V_{an} = -\frac{2V_s}{3} \qquad V_{ab} = -V_s$$
$$V_{bn} = \frac{V_s}{3} \qquad V_{bc} = 0$$
$$V_{cn} = \frac{V_s}{3} \qquad V_{ca} = -V_s$$

300° – 360° (S_4, S_5 and S_6 are ON)



$$V_{an} = \frac{-V_s}{3} \qquad V_{ab} = 0$$
$$V_{bn} = \frac{-V_s}{3} \qquad V_{bc} = -V_s$$
$$V_{cn} = \frac{2V_s}{3} \qquad V_{ca} = V_s$$

b)

Merits of 120° Conduction Mode:

Lower Harmonics: 120° conduction mode inverters produce fewer harmonics in the output waveform compared to 180° mode inverters. This leads to a cleaner output voltage and reduced distortion.

Improved Efficiency: Due to reduced switching losses, 120° mode inverters can be more efficient, especially at lower power levels.

Smaller Heat Sink Requirements: Lower switching losses result in reduced heat generation, allowing for smaller and more cost-effective heat sinks.

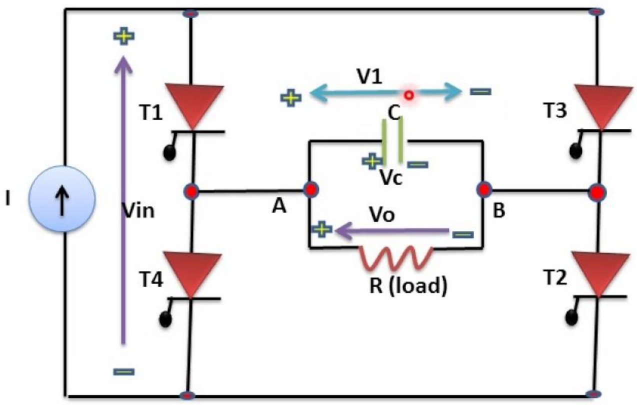
Demerits of 120° Conduction Mode:

Limited Voltage Output Range: 120° conduction mode inverters have a more limited voltage output range compared to 180° mode inverters. This limitation can be a disadvantage in applications where a wide voltage range is required.

Reduced Voltage Utilization: The output voltage of a 120° mode inverter is limited to around 0.9 times the DC bus voltage, which means it may not fully utilize the available DC voltage, resulting in lower output power.

Reduced Control Flexibility: In some applications, precise control of the output voltage may be challenging with 120° conduction mode inverters due to their limited voltage range.

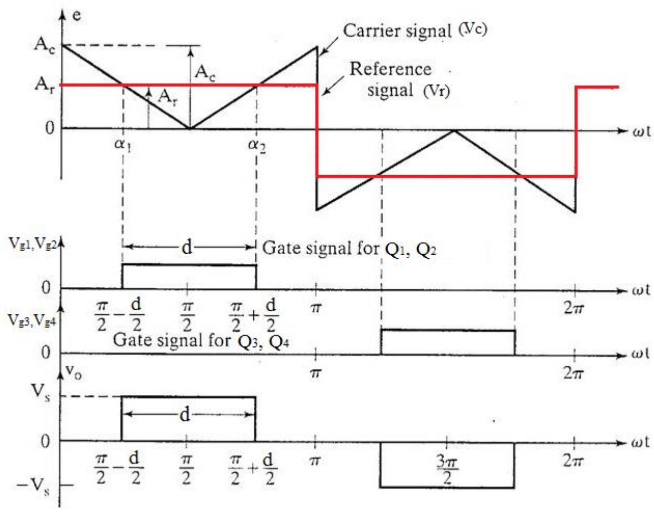
Q16	a)	Explain the operation of single phase capacitor commutated inverter with relevant waveforms. (Marks7)
	b)	Describe single pulse width modulation used in inverters (7 Marks)
Ans	a)	



b)

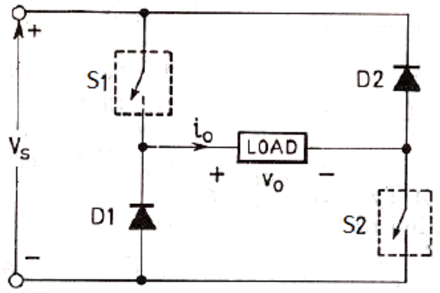
In single pulse width modulation technique, there is only one pulse per half cycle and the output voltage is changed by varying the width of the pulse.

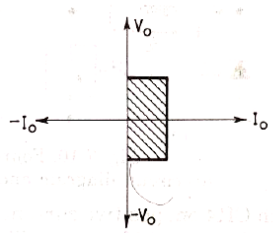
The gating signals are generated by comparing the rectangular control signal (reference signal) V_r with triangular carrier signal V_c.



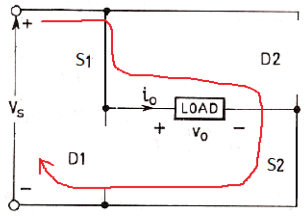
Module 4

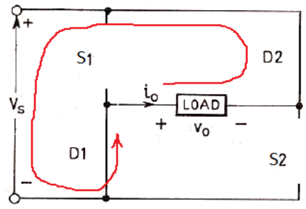
Q17	a) Explain the Two quadrant type class D chopper operation in different quadrants with relevant output voltage and output current waveforms. (8 Marks) b) Explain the pulse width modulation techniques in DC-DC converter. (6 Marks)
Ans	a) Class D Chopper • It can operate in first and fourth quadrants.





Quadrant – 1 operation

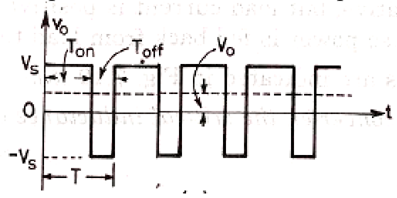


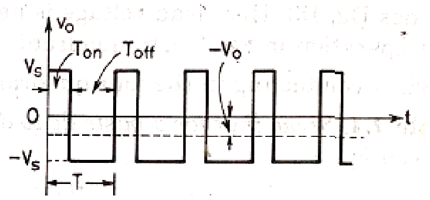


- When switch S1 and S2 are ON, source will be connected across the load and current flows in a direction shown in the figure and $V_o = V_s$.
- When switches are turned OFF, load current continues the flow in the same direction through the diodes D1 and D2 and back to the source V_s.
  Thus, $V_o = -V_s$.
- That is, output voltage can be positive as well as negative whereas output current is always positive and thus the quadrants of operation are first and fourth. quadrant.

Quadrant – 4 operation

- When ON period of the switches are more average output voltage will be positive and thus operation will be in first quadrant.
- When OFF period is more, average output voltage will be negative and thus operation will be in fourth quadrant.



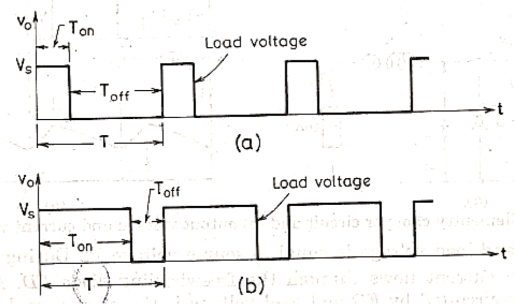


b) **Time Ratio Control:** Here duty ratio is varied, to control the output voltage of the chopper. This can be achieved in 2 ways.

   a) **Constant frequency control (PWM control)**
   b) **Variable frequency control (Frequency Modulation control)**

<u>**Constant frequency control (PWM control)**</u>

In this method, ON-period of the switch is varied by keeping switching frequency constant.



Q18

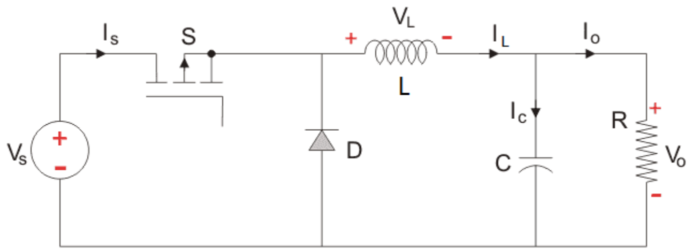
a) Explain with neat circuit diagram and waveforms, the operation of Buck Regulator. Derive the expression for its output voltage. (9 Marks)

b) In a step down chopper the DC input voltage is 100V. The MOSFET switch is having a switching frequency of 2kHz. Find the duty cycle and average dc output voltage if the turn on period of switch is 0.2ms. (5 Marks)

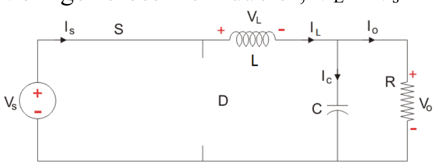
Ans

a) Buck Regulator

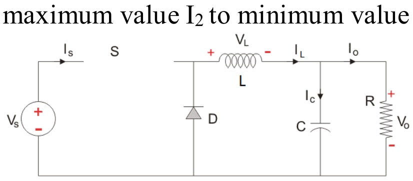
- Buck converter is a voltage step-down converter.
- Output voltage will be less than the input voltage.

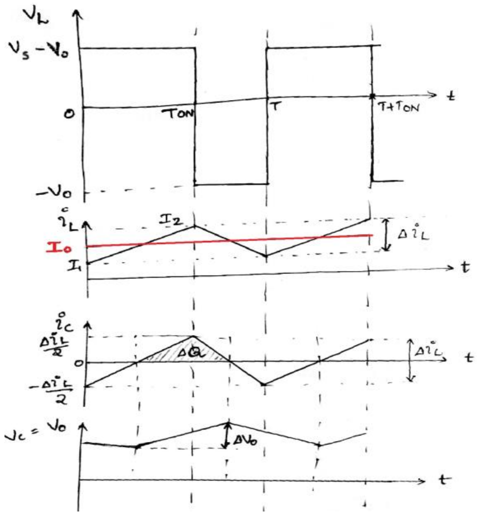


- When switch S is ON, diode D will be reverse biased and it will not conduct.
- Inductor L, stores energy and inductor current I_L will increase from a minimum value I_1 to a maximum value I_2.
  Voltage across the inductor, $V_L = V_s - V_o$



- When switch S is turned OFF, inductor current finds a path through diode D.
- Voltage across the inductor, $V_L = -V_o$
- Since voltage across the inductor is negative, inductor current I_L decreases from maximum value I_2 to minimum value I_1.





Relation between Output Voltage & Input Voltage

At steady state, **the sum of energy stored in an inductor and energy released by an inductor will be zero** **(Volt-Second Balance Rule)**

$$Energy = Voltage \times Current \times Time$$

$$(V_s - V_o)\left(\frac{I_1 + I_2}{2}\right)T_{ON} + -V_o\left(\frac{I_1 + I_2}{2}\right)T_{OFF} = 0$$

$$(V_s - V_o)\left(\frac{I_1 + I_2}{2}\right)T_{ON} = V_o\left(\frac{I_1 + I_2}{2}\right)T_{OFF}$$

$$(V_s - V_o)T_{ON} = V_o T_{OFF}$$

$$V_s T_{ON} = V_o(T_{OFF} + T_{ON}) = V_o T$$

$$V_o = V_s \frac{T_{ON}}{T}$$

$$\boxed{V_o = DV_s}$$

Where,
D is **duty ratio or duty cycle**

b)

Given,

$V_{in} = 100\ V$

$f = 2\ kHz = 2000\ Hz$

$T_{ON} = 0.2\ ms = 0.2 \times 10^{-3}\ s$

Period, $T = \frac{1}{f} = \frac{1}{2000} = \underline{\underline{0.5\ ms}}$

Duty cycle, $D = \frac{T_{ON}}{T}$

$= \frac{0.2}{0.5} = \underline{\underline{0.4}}$

		$V_0 = DV_{in}$ $= 0.4 \times 100 = \underline{\underline{40\,V}}$
		Module 5
	Q19	Describe the speed control of DC motor in simultaneous and non-simultaneous conduction mode using single phase Dual Converter. (14 Marks)
	Ans	Speed control of a DC motor using a single-phase dual converter involves controlling the voltage and polarity applied to the armature of the DC motor. The dual converter is a type of power electronic converter that can control both the magnitude and polarity of the DC voltage supplied to the motor. The control can be achieved in two modes: simultaneous conduction mode and non-simultaneous conduction mode. **1. Simultaneous Conduction Mode:** In simultaneous conduction mode, both the upper and lower thyristors (or other switching devices) of the dual converter conduct at the same time. This mode allows for bidirectional power flow through the motor, meaning the motor can run in both forward and reverse directions. Here's how speed control is achieved in this mode: • When both the upper and lower thyristors conduct simultaneously, the full DC voltage is applied to the motor armature terminals without any change in polarity. This results in the maximum motor speed in one direction (either forward or reverse) depending on the armature current direction. • To control the motor speed, a phase-controlled firing angle control technique is used. By adjusting the firing angle of the thyristors in both the upper and lower arms of the dual converter, the effective voltage applied to the motor can be varied, allowing for speed control. • Increasing the firing angle (delaying the thyristor firing) reduces the effective voltage, slowing down the motor, while decreasing the firing angle (advancing the thyristor firing) increases the voltage, speeding up the motor. • The simultaneous conduction mode is suitable for applications where bi-directional speed control is required. **2. Non-Simultaneous Conduction Mode:** In the non-simultaneous conduction mode, the upper and lower thyristors of the dual converter do not conduct at the same time. This mode allows for unidirectional power flow through the motor, meaning the motor can run in either the forward or reverse direction, but not both simultaneously. Here's how speed control is achieved in this mode: • To change the direction of the motor, the upper and lower thyristors are controlled independently. When one set of thyristors conducts (e.g., upper thyristors), the voltage is applied in one direction, and when the other set conducts (e.g., lower thyristors), the voltage is applied in the opposite direction. • Speed control is achieved by controlling the firing angle of the conducting set of thyristors. Increasing the firing angle reduces the effective voltage, slowing down

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the motor, while decreasing the firing angle increases the voltage, speeding up the motor.

- The non-simultaneous conduction mode is suitable for applications where unidirectional speed control is sufficient, and reversing the motor direction is not frequently required.

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<tr><td></td><td>Q20</td><td>

a) Explain the working of Four quadrant chopper drives with neat circuit diagram. (6 Marks)

b) Explain the stator voltage control of three phase induction motor drive. (8 Marks)

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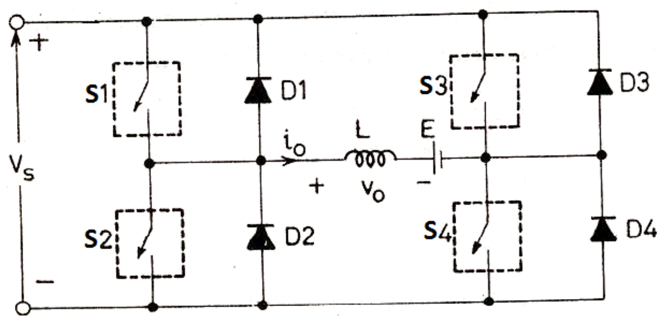
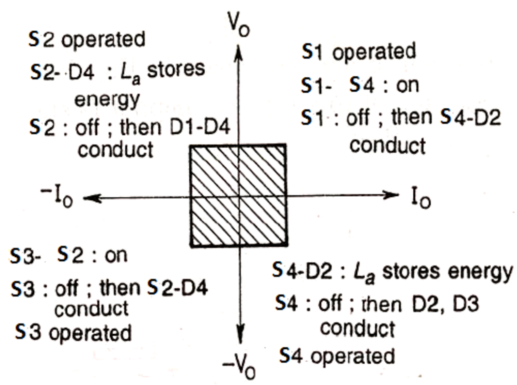
<tr><td></td><td>Ans</td><td>

a)





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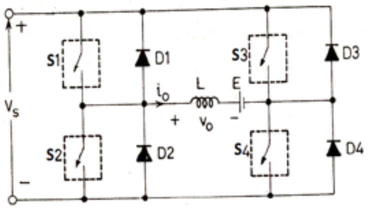
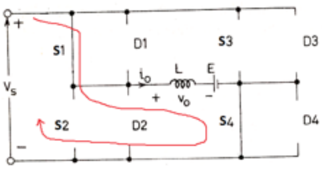
Quadrant – 1

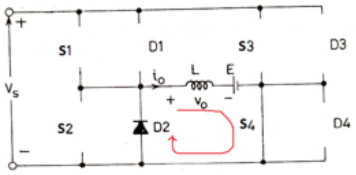


- Switch S1 is operated
- Switch S4 is always kept ON.
- When S1 is ON, load will be connected across the source through S1 and S4 and thus inductor stores energy. Current flow is in the marked direction and $V_o = V_s$.
- When S1 is turned OFF, load current continues the same direction through S4 and D2. Thus $V_o = 0$
- Since both output voltage and current are positive, first quadrant of operation

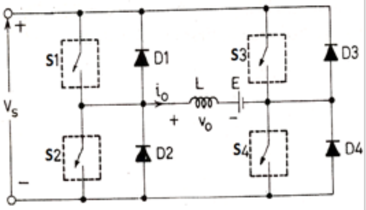
When Switch S1 is ON



When Switch S1 is OFF

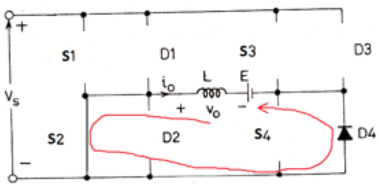


Quadrant – 2

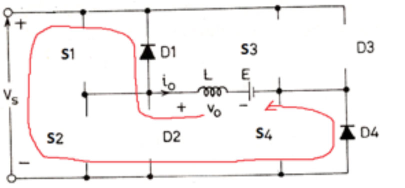


- S[illegible]rated
- When S2 is ON, inductor L stores energy from the source E through diode D4. Current flow is opposite to the marked direction and $V_o = 0$.
- When S2 is turned OFF, load current continues the same direction through D1, source Vs and D4. Thus $V_o = V_S$
- That is voltage is positive whereas current is negative. Thus second quadrant of operation

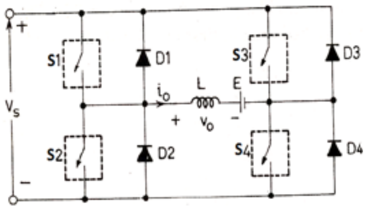
When Switch S2 is ON



When Switch S2 is OFF

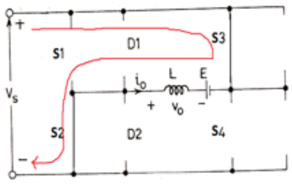


Quadrant – 3

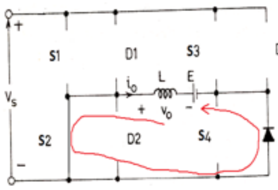


- Switch S3 is operated.
- Switch S2 is always kept ON.
- When S3 is ON, load will be connected across the source through S3 and S2 and thus inductor stores energy. Current flow is opposite to the marked direction and $V_o = -V_s$.
- When S3 is turned OFF, load current continues the same direction through S2 and D4. Thus $V_o = 0$
- Since both output voltage and current are negative, third quadrant of operation

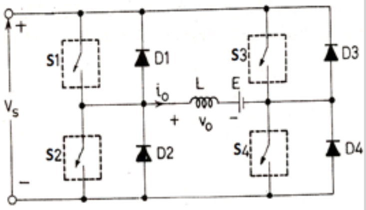
When Switch S3 is ON



When Switch S3 is OFF

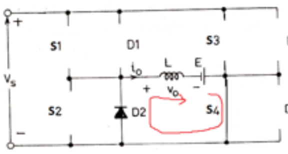


Quadrant – 4

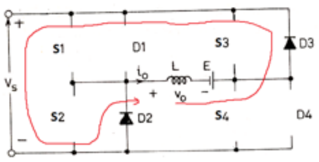


- Load emf E should be reversed to achieve operation in fourth quadrant.
- Switch S4 is operated
- When S4 is ON, inductor L stores energy from the source E through S4 and diode D2. Current flow is in the marked direction and $V_o = 0$.
- When S4 is turned OFF, load current continues the same direction through D3, source Vs and D2. Thus $V_o = -V_S$
- Since output voltage is negative and current is positive, operation is in fourth quadrant.

When Switch S4 is ON



When Switch S4 is OFF



b) Stator voltage control is a common method used to control the speed of a three-phase induction motor drive. It involves varying the voltage applied to the stator windings of the motor while maintaining a constant frequency. This method is often used in applications where precise control of motor speed is required. Here's an explanation of stator voltage control for a three-phase induction motor drive:

Basic Principle:

In a three-phase induction motor, the speed of rotation (synchronous speed) is directly proportional to the frequency of the applied voltage and inversely proportional to the number of poles. The fundamental equation is: Ns = 120 * f / P where:

Ns = Synchronous speed (in RPM)

		f = Frequency of the applied voltage (in Hertz) P = Number of motor poles By maintaining a constant frequency and varying the voltage applied to the stator, you can control the motor's speed. Increasing the voltage above the rated value will increase the motor speed, while decreasing it below the rated value will decrease the speed.